SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **K.Sasidhar Reddy(192525140)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of the student:**K.Sasidhar Reddy*

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

A) Aim

To solve the Water Jug Problem using Artificial Intelligence by obtaining exactly 2 gallons of water in the 4-gallon jug with the help of a 3-gallon jug.

Problem Description

Two jugs are available: one has a capacity of 4 gallons and the other has a capacity of 3 gallons. Neither jug has any measurement markings. A water pump is available to fill the jugs, and water can be poured from one jug to another or emptied onto the ground. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug without using any additional measuring devices.

Assumptions

1. The 4-gallon jug can hold a maximum of 4 gallons.
2. The 3-gallon jug can hold a maximum of 3 gallons.
3. Either jug can be filled completely using the pump.
4. Water can be emptied onto the ground.
5. Water can be transferred between the two jugs until the source jug is empty or the destination jug is full.
6. No other measuring devices are available.

State-Space Representation

In Artificial Intelligence, a state represents the amount of water present in both jugs.

- **State:** (x, y), where x is the amount of water in the 4-gallon jug ($0 \leq x \leq 4$) and y is the amount of water in the 3-gallon jug ($0 \leq y \leq 3$).
- **Initial State:** (0, 0)
- **Goal State:** (2, y) for any y, i.e. exactly 2 gallons in the 4-gallon jug. In this solution the goal is reached at (2, 3).
- **Operators:** Fill a jug, empty a jug, or pour water from one jug into the other.

Production Rules

Rule	Condition	Action
1	(x, y) and $x < 4$	(4, y) – fill the 4-gallon jug
2	(x, y) and $y < 3$	(x, 3) – fill the 3-gallon jug
3	(x, y) and $x > 0$	(0, y) – empty the 4-gallon jug
4	(x, y) and $y > 0$	(x, 0) – empty the 3-gallon jug
5	(x, y), $x + y \geq 3$, $x > 0$	$(x - (3 - y), 3)$ – pour 4-gal into 3-gal until 3-gal is full

Rule	Condition	Action
6	(x, y), $x + y \geq 4$, $y > 0$	(4, $y - (4 - x)$) – pour 3-gal into 4-gal until 4-gal is full
7	(x, y), $x + y \leq 3$, $x > 0$	(0, $x + y$) – pour all of 4-gal into 3-gal
8	(x, y), $x + y \leq 4$, $y > 0$	($x + y$, 0) – pour all of 3-gal into 4-gal

Step-by-Step Solution

Step	Action	4-Gallon Jug (x)	3-Gallon Jug (y)	Rule
0	Start with both jugs empty	0	0	–
1	Fill the 4-gallon jug completely	4	0	1
2	Pour water from the 4-gallon jug into the 3-gallon jug until it is full	1	3	5
3	Empty the 3-gallon jug onto the ground	1	0	4
4	Transfer the remaining water of the 4-gallon jug into the 3-gallon jug	0	1	7
5	Fill the 4-gallon jug completely again	4	1	1
6	Pour water from the 4-gallon jug into the 3-gallon jug until it is full	2	3	5

Solution path: (0,0) → (4,0) → (1,3) → (1,0) → (0,1) → (4,1) → (2,3)

Algorithm

1. Start with both jugs empty, (0, 0).
2. Fill the 4-gallon jug.
3. Pour water into the 3-gallon jug until it is full.
4. Empty the 3-gallon jug.
5. Transfer the remaining water of the 4-gallon jug into the 3-gallon jug.
6. Fill the 4-gallon jug again.
7. Pour water into the 3-gallon jug until it is full.
8. Stop when the 4-gallon jug contains exactly 2 gallons.

Result

The Water Jug Problem is successfully solved using the state-space search technique in Artificial Intelligence. By applying the operations of filling, emptying, and transferring water in 6 steps, exactly 2 gallons of water are obtained in the 4-gallon jug without using any additional measuring devices.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percepts for this agent?

ii. Characterize the operating environment.

iii. What are the actions the agent can take?

iv. How can one evaluate the performance of the agent?

v. What sort of agent architecture do you think is most suitable for this agent? Why?

A) i. What are the percepts for this agent?

Percepts are the information collected by the Mars Rover through its sensors.

- **Camera:** Captures images of the Martian surface.
- **Proximity sensors:** Detect rocks and obstacles.
- **Navigation sensors:** Identify the rover's position and direction.
- **Scientific instruments:** Analyse soil, rocks, and other samples.
- **Environmental sensors:** Measure temperature, pressure, and radiation.

ii. Characterize the operating environment.

The Mars Rover operates in the following environment:

- **Partially observable:** It cannot observe the entire Martian surface at once.
- **Single-agent:** It performs exploration tasks independently, although ground controllers may provide commands.
- **Stochastic:** Unexpected obstacles and environmental conditions may affect its actions.
- **Sequential:** Each action affects future decisions and movements.
- **Dynamic:** Environmental conditions can change during exploration.
- **Continuous:** Movement, time, temperature, and distance involve continuous values.
- **Unknown:** The rover must explore unfamiliar terrain and learn about its surroundings.

iii. What are the actions the agent can take?

The Mars Rover can perform the following actions:

1. Move forward, backward, and turn.
2. Capture images of the surroundings.
3. Detect and avoid obstacles.
4. Collect soil and rock samples.
5. Analyse samples using scientific instruments.
6. Send collected data and images to Earth.
7. Select suitable locations for further exploration.
8. Recharge or conserve energy when required.

iv. How can one evaluate the performance of the agent?

The performance of the Mars Rover can be evaluated using the following criteria:

- **Scientific discovery:** Number and quality of useful findings.

- **Navigation accuracy:** Ability to reach target locations safely.
- **Sample analysis:** Accuracy and usefulness of sample results.
- **Energy efficiency:** Effective use of available battery power.
- **Safety:** Ability to avoid obstacles and prevent damage.
- **Mission completion:** Successful completion of exploration objectives.
- **Communication:** Successful transmission of collected data to Earth.

v. What sort of agent architecture is most suitable for this agent? Why?

A model-based, goal-based, and utility-based agent architecture, with learning capability, is most suitable for the Mars Rover.

- **Model-based agent:** Maintains an internal map of the terrain and remembers observed obstacles.
- **Goal-based agent:** Plans actions to reach target locations and collect samples.
- **Utility-based agent:** Chooses actions that balance scientific value, safety, time, and energy consumption.
- **Learning agent:** Improves its decisions using previous observations and exploration results.

Reason: The Mars Rover operates in a partially observable, unknown environment with limited energy and delayed communication with Earth. Combining these architectures helps it make intelligent decisions, navigate safely, analyse samples, and achieve its scientific goals efficiently.

Conclusion

The Mars Rover is an intelligent agent that uses sensors to perceive its environment and actuators to perform actions. A combination of model-based, goal-based, utility-based, and learning-agent capabilities enables it to explore Mars, analyse samples, avoid obstacles, and achieve mission objectives successfully.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

A) Problem Components

1. Initial State: An empty 8×8 chessboard (no queens placed).
2. State Space: All arrangements of 0 to 8 queens on the board. Using the incremental formulation, only arrangements in which no two queens attack each other are considered.
3. Actions: Place a queen in a safe square (not attacked by any queen already placed) in the leftmost empty column.
4. Transition Model: Returns the board with one more queen added at the selected square.
5. Goal Test: All 8 queens are placed without any two sharing the same row, column, or diagonal.

6. Path Cost: Each queen placement has a cost of 1 (so every solution has path cost 8; the solution itself matters, not the path).

Search Strategies

- **Breadth-First Search (BFS):** Explores states level by level (all boards with k queens before boards with k + 1 queens). It finds a solution but needs a very large amount of memory.
- **Depth-First Search (DFS):** Places queens one by one in successive columns, going as deep as possible, and backs up when a conflict occurs. Memory needed is small.
- **Backtracking:** Removes the last queen when no safe position is available in the current column and tries the next position in the previous column.

One Valid Solution

The following row positions represent the queens in columns 1 to 8:

Column	1	2	3	4	5	6	7	8
Row	1	5	8	6	3	7	2	4

Queens are at (row, column): (1,1), (5,2), (8,3), (6,4), (3,5), (7,6), (2,7), (4,8). All rows are different, all columns are different, and no two queens lie on the same diagonal, so the goal test is satisfied.

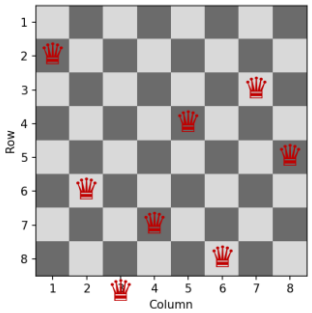


Figure: A valid 8-Queens solution

Result

The 8-Queens problem can be solved using DFS with backtracking, ensuring that no two queens share the same row, column, or diagonal.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

A) Type of Problem-Solving Agent

A Goal-Based Agent is suitable because the customer wants to reach a specific destination by selecting a suitable cab type. The agent formulates the goal (be at the destination), formulates the problem, searches for a suitable cab and route, and then executes the booking.

Problem Formulation

- **Initial State:** Customer at the source location.
- **Goal State:** Customer reached the destination.
- **Actions:** Select cab type (Mini, Micro, Sedan, Shared, Prime), confirm booking, ride to destination.
- **Path Cost:** Fare and travel time.

Pseudocode

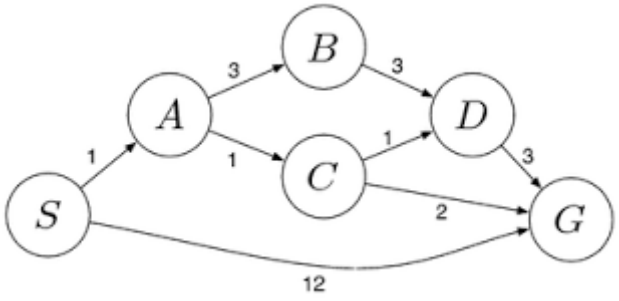
START
Input source location
Input destination location
Display cab types: Mini, Micro, Sedan, Shared, Prime
Select cab type
Calculate fare and distance
Display estimated arrival time
Confirm cab booking
Assign available cab
Travel to destination
IF destination reached THEN
  Display trip completion
STOP

Result

The OLA Cab Booking system uses a Goal-Based Agent to help customers select a cab and reach their destination. It considers the source, destination, cab type, fare, and availability to complete the booking.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Edges (cost): S→A (1), S→G (12), A→B (3), A→C (1), B→D (3), C→D (1), C→G (2), D→G (3).

A) Aim

To find the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search algorithm.

Algorithm

1. Start from node S with cost 0.
2. Insert S into the priority queue.
3. Select (remove) the node with the lowest cumulative cost.
4. Expand the node and calculate the costs of its neighboring nodes.
5. Update the priority queue with the new paths if their costs are lower.
6. Repeat until the destination G is selected for expansion.
7. Display the least-cost path and its total cost.

Pseudocode

```
START
Insert S into priority queue with cost 0
WHILE priority queue is not empty:
   Remove node with lowest cost
   IF node is G:
      Display path and total cost
      STOP
   FOR each neighbor of node:
      Calculate new cost = cost(node) + edge cost
      IF neighbor is not visited OR new cost is lower:
         Insert / update neighbor in priority queue
STOP
```

Step-by-Step Execution

Step	Node expanded (cost)	New paths generated (cumulative cost)	Priority queue after step (ordered by cost)
1	S (0)	S→A = 1; S→G = 12	A(1), G(12)
2	A (1)	S-A→B = 1+3 = 4; S-A→C = 1+1 = 2	C(2), B(4), G(12)
3	C (2)	S-A-C→D = 2+1 = 3; S-A-C→G = 2+2 = 4 (better than 12, so G updated)	D(3), B(4), G(4)
4	D (3)	S-A-C-D→G = 3+3 = 6 (not better than 4, discarded)	B(4), G(4)
5	G (4)	G selected – goal reached	–

Paths to G found by UCS: S→G = 12; S→A→C→D→G = 6; S→A→C→G = 4 (minimum). Node B (cost 4) is never needed, since its only route to G, B→D→G, costs 4 + 3 + 3 = 10.

Result

Uniform Cost Search finds the least-cost path from S to G by expanding the path with the lowest cumulative cost first. The least-cost path is **S → A → C → G** with a total cost of **4** units.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem